MUGESH S

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate and Full Stack .NET Developer with internship experience and Accenture full stack training. Skilled in building scalable, secure web applications using C#, ASP.NET Core, Web API, Entity Framework Core, SQL Server and ReactJS. Experience with RESTful microservices, modular code design, Microsoft Azure, Docker, Kubernetes and DevOps practices, plus Generative AI and RAG integration. IEEE-published researcher and Fortinet Certified Professional (FCP) focused on writing maintainable, security-first code.

TECHNICAL SKILLS

- **Programming Languages:** C#, Java, Python, C++, JavaScript
- **.NET Development:** .NET, ASP.NET Core MVC, ASP.NET Core Web API, Entity Framework Core, RESTful APIs, Microservices
- **Web Development:** ReactJS, HTML, CSS, JavaScript, Bootstrap, Spring Boot
- **Databases:** SQL Server, MySQL, MongoDB
- **Cloud & DevOps:** Microsoft Azure, Docker, Kubernetes, DevOps, Git, Linux
- **AI Tools & Integration:** Generative AI, RAG, Azure OpenAI, GitHub Copilot
- **Software Engineering:** Object-Oriented Programming, Modular Design, Debugging, Unit Testing
- **Security:** Secure Authentication, Cryptography, Snort, Wireshark

EXPERIENCE

Accenture – .NET Full Stack Development Training May 2026 – Sep 2026

- Completed full stack .NET training covering C#, ASP.NET Core MVC, Web APIs and microservices architecture.
- Designed databases and implemented data access using SQL Server and Entity Framework Core.
- Built interactive front ends using HTML, CSS, JavaScript and React.
- Gained exposure to Microsoft Azure, Docker, Kubernetes and DevOps practices for deployment and management.
- Applied concepts through hands-on exercises, case studies and a capstone full stack application.

Mist Software Solutions Private Limited – Software Development Intern May 2025

- Developed and maintained full stack web application features using JavaScript, HTML, CSS, Bootstrap and MySQL.
- Designed modular code and reusable components to improve maintainability.
- Implemented and tested backend logic for authentication, form validation and database operations.
- Debugged issues by analysing logs, reproducing bugs and delivering fixes within deadlines.
- Collaborated with team members through code reviews and iterative feedback.

PROJECTS

Hybrid Post-Quantum Cryptography–Steganography Framework Mar 2026

- Built a multi-layered multimedia security system integrating McEliece KEM, adaptive RC6 encryption and DCT–IDCT O-Pattern LSB steganography to protect text, image and audio transmissions in medical IoT environments.
- Evaluated resilience against noise, compression and geometric attacks, achieving PSNR above 39 dB with low MSE and high SSIM, within a scalable, server-backed key management architecture.

JournalProbe – Full Stack Web Application (React) Jul 2025

- Built a React web application with user authentication, protected routes and a responsive user interface.
- Implemented reusable components and state management for maintainability and scalability.
- Added bilingual support (English and Tamil) to improve accessibility.
- Applied good practices for code structure, error handling and frontend security.

Secure Authentication System Simulation (Python) Aug 2025

- Designed and implemented a Kerberos-based authentication workflow in Python.
- Applied encryption, ticket-based access control and validation logic.
- Analysed system behaviour under replay and impersonation attack scenarios.
- Documented design decisions and compared results with modern authentication frameworks.

EDUCATION

Bachelor of Engineering – Computer Science Engineering Oct 2022 – Mar 2026
Sri Krishna College of Technology CGPA: 7.54

CERTIFICATIONS

- Fortinet Certified Professional (FCP) Sep 2025
- Fortinet Certified Associate in Cybersecurity Jun 2025

PUBLICATIONS

- **A Hybrid Post-Quantum Secure Framework Integrating McEliece KEM, RC6 Encryption and Steganography for IoT Data Protection** – Suma Sira Jacob, Manoj Shanmugam, Keerthana Pandiyan, Mugesh Selvakumar. 2025 International Conference on Intelligent Innovations in Engineering and Technology (ICIIET), IEEE, 2025. DOI: 10.1109/ICIIET65921.2025.11378708